Classifying Clinical Notes into Medical Specialties

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1 Introduction

The task is to assign one of 12 medical specialties to a clinical record (description, sample name, transcription and keywords). We found that the first keyword almost always names the specialty, but about one record in five has no keywords; there, the system must read the clinical text.

Our research question is: **how much of the task can be solved from the clinical text alone, and can data augmentation make a model that uses keywords robust to records in which they are missing?** We compare text-only and keyword-based models, also on records without keywords, and design an augmentation that simulates missing keywords.

2 Data

2.1 Data Analysis

Of the 2,616 training records, three with transcription text in the label field (a parsing error) were removed, leaving 2,613; the official test set has 398. The classes are imbalanced: Surgery has 876 records (33.5%) and Dermatology 25, a ratio of 35 to 1 (Table 2). Median lengths: transcription 387 words, description 16.

(O1) The first keyword reveals the label. In 2,171 of the 2,172 training records with keywords, the first keyword is the specialty name in lower case (e.g., “cardiovascular / pulmonary”). Keywords are missing in 441 training records (16.9%) and 78 test records (19.6%). They are not a random sample: 27 of the 40 Psychiatry-Psychology records lack keywords, but only 3 of the 71 Ophthalmology ones.

(O2) The same note appears under several labels. There are only 1,817 distinct transcriptions; 702 occur more than once, and in all but one case the copies have different labels, mostly Orthopedic/Surgery (118 notes), Cardiovascular-Pulmonary/Surgery (113) and Gastroenterology/Surgery (95). 526 of the 876 Surgery records are such copies. Hence, no transcription-only model can exceed 70.3% training accuracy.

(O3) Fields are sometimes misplaced. 30 records have an empty transcription, sometimes because the note was placed in the description (up to 4,933 words). Sections are joined by commas without spaces (“,PLAN:”). Also, 239 of the 398 test transcriptions occur in the training set; we did not use a look-up, which would often return conflicting labels.

2.2 Data Augmentation

Both are applied only to the training part of each fold. **(A1) Keyword dropout:** for each record with keywords we add a copy with all keyword fields emptied (2,172 copies). It simulates test records without keywords (O1), forcing a keyword-based model to also learn from the text. **(A2) Minority oversampling:** classes with fewer than 150 records (Neurosurgery, Ophthalmology, Psychiatry-Psychology, Dermatology) are increased to 150 with perturbed copies in which each sentence is deleted with probability 0.2 and sentences are shuffled

within windows of three. This adapts EDA's random deletion and swap [Wei and Zou, 2019] to sentences, keeping clinical terms intact (401 copies). We also test A1+A2.

3 Models

We used scikit-learn [Pedregosa et al., 2011]. Except in M0, each field has its own TF-IDF (Term Frequency–Inverse Document Frequency) vectorizer, so a word gets different weights in different fields. Tokens are lower-cased alphabetic words (2+ letters) with log-scaled term frequency; we kept stop words and did not stem, since words like "left" or "no" are clinically relevant. Fields: description and sample name (1–2-grams), transcription (1–2-grams in at least 2 and at most 90% of documents; commas before section names replaced by spaces, O3), and keywords *except the first* (unigrams).

M0 – Baseline: Multinomial Naive Bayes on TF-IDF of the lower-cased, Porter-stemmed [Porter, 1980] description, as in the project statement. **M1 – Text-only Logistic Regression:** one-vs-rest, regularisation $C = 10$ (C is the inverse of the penalty strength) and balanced class weights (each class weighted inversely to its frequency). **M2 – Text-only linear SVM** [Joachims, 1998]: $C = 0.5$, balanced class weights. **M3 – Joint text + keyword SVM:** M2 plus a one-hot first keyword (empty is a category), scaled by 3 so it is not drowned by the text features. **M4 – Hybrid rule + SVM:** if the first keyword is one of the 12 specialty names, return it; otherwise return the prediction of M2.

M1–M2 measure what the text alone achieves; M3–M4 combine text and keywords jointly or by a rule. A joint model may become keyword-dependent, which A1 aims to prevent.

4 Experimental Setup

The official test set only produces `results.txt`. We use 5-fold cross-validation on the training records, stratified by label and **grouped by transcription**, so copies of a note (O2) never appear in both training and test data. Each held-out fold (522–523 records) is one of our own test sets. We report accuracy, macro-F1 (unweighted mean of per-class F1) and minimum per-class F1, averaged over folds, plus macro-F1 on the 441 records without keywords ("no-kw").

Model	Aug.	Acc	M-F1	Min-F1	no-kw
M0	–	.396	.131	.000	.092
M1	–	.586	.584	.354	.570
M2	–	.591	.612	.405	.621
M2	A2	.590	.614	.417	.614
M3	–	.934	.924	.845	.496
M3	A1	.941	.927	.822	.627
M4	–	.942	.930	.842	.621
M4	**A1**	**.942**	**.930**	**.845**	**.623**

Table 1: Mean of 5 grouped folds. M-F1: macro-F1; Min-F1: lowest class F1; no-kw: macro-F1 without keywords.

4.1 Parameters/Hyperparameters

We tested a small grid on the same folds (Appendix A, Table 4): C for M1/M2, minimum n-gram document frequency, M3 keyword weight, and A2 size and deletion rate. No change exceeded the fold standard deviation of macro-F1 (0.02), so we kept the initial values (scores may be slightly optimistic, as selection and reporting share folds). Seed: 42.

5 Results

Table 1 shows the main results (all combinations in Appendix A, Table 3); fold standard deviation is 0.01 (accuracy) and 0.02 (macro-F1). From text alone, M2 reaches 0.591 accuracy and 0.612 macro-F1, far above the baseline, which never predicts 6 of the 12 classes. Adding the keyword raises accuracy to 0.94. Without augmentation, the joint M3 drops to 0.496 macro-F1 on records without keywords, below the text-only models; with keyword dropout (A1) it reaches 0.627. A2 had no effect beyond fold variation. Our final system is **M4 with A1**: it ties with the

Class	N	P	R	F1	M2
Cardio-Pulm.	299	.902	.957	.929	.618
Dermatology	25	1.00	.880	.936	.571
Gastroent.	192	.968	.943	.955	.590
Gen. Medicine	207	.857	.928	.891	.769
Neurology	181	.902	.917	.910	.607
Neurosurgery	78	.811	.936	.869	.418
Obst.-Gyn.	135	.926	.933	.930	.617
Ophthalm.	71	.986	1.00	.993	.712
Orthopedic	292	.949	.952	.950	.573
Psychiatry	40	.841	.925	.881	.818
Radiology	217	.981	.931	.955	.495
Surgery	876	.988	.945	.966	.536

Table 2: Per-class P, R and F1 of M4+A1; F1 of text-only M2 (A1+A2).

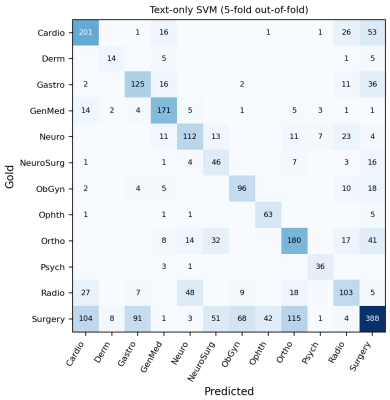


Figure 1: Confusion matrix of text-only M2 (A1+A2), out-of-fold.

best, has the highest minimum F1 and is interpretable. Table 2 gives its per-class results. Figure 1 shows the confusion matrix of the text-only M2, as M4 errs on the same records (M4's is in Appendix A).

6 Discussion

The final system makes 151 errors, and 150 of them are on records without keywords. On these records, accuracy is 0.80 when the note is unique, but only 0.44 when the same note also appears with another label (171 records). In Figure 1, Surgery has the lowest recall (0.44), and most of its errors go to the specialties it shares notes with. Radiology/Neurology and Orthopedic/Neurosurgery are also confused.

Example 1 (gold Surgery, predicted Cardiovascular-Pulmonary). "PROCEDURE: Left heart catheterization, left ventriculography. . . " The same note appears in the data labelled Cardiovascular-Pulmonary. Both labels are correct in our view, but no single-label model can get both identical copies right. 20 errors are like this.

Example 2 (gold Radiology, predicted Neurology). The sample name is "CT Brain – Stroke", but the note begins "CC: Falls. HX: This 51y/o RHF fell four times. . . ", a neurological case history with the imaging at the end. The label comes from the exam in the title; the model follows the content, as in all nine such errors.

Example 3 (gold Gastroenterology, predicted General Medicine). "REASON FOR CONSULT: I was asked to see the patient for C. diff colitis. . . . " This note has no duplicate, so the model is simply wrong. Most of the note follows the usual consultation structure (history, medications, review of systems), common in General Medicine. In a bag-of-words these words outweigh the reason for consult, where the specialty is stated.

Answer to the research question. With the text alone we reach about 59% accuracy, and the limit is around 70%, because many labels depend on how a note was filed and not on what it says (O2). Keywords close most of this gap. But a model learning from both relies too much on the keywords unless trained with keyword dropout. A1 worked because it reproduces a real test-set situation, whereas A2 only varies notes the model has already seen. **Limitations.** Labels were not corrected, as required. Results without keywords rely on 441 records, so small differences are not reliable.

7 Future Work

We would (i) compare M2 with a fine-tuned ClinicalBERT; (ii) add section-based features that give more weight to the reason for consult (Example 3); and (iii) treat Surgery as an extra label (multi-label, Example 1).

References

[Joachims, 1998] Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. In *Proceedings of ECML*, pages 137–142.

[Pedregosa et al., 2011] Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

[Porter, 1980] Porter, M. F. (1980). An algorithm for suffix stripping. *Program*, 14(3):130–137.

[Wei and Zou, 2019] Wei, J. and Zou, K. (2019). EDA: Easy data augmentation techniques for boosting performance on text classification tasks. In *Proceedings of EMNLP-IJCNLP*, pages 6382–6388.

Use of LLMs. Claude (Anthropic) was used to assist in the exploratory data analysis, in the implementation of the pipeline and in a first draft of this paper. All code was executed and checked by the group, and all reported numbers and examples come from our own runs and our own reading of the data.
Dataset: adapted from the Medical Transcriptions dataset on Kaggle (`https://www.kaggle.com/datasets/tboyle10/medicaltranscriptions`), originally scraped from `https://mtsamples.com`.

Appendix A: Extra Figures and Tables

Model	Augmentation	Accuracy	Macro-F1	Min-F1	Macro-F1 no-kw
M0 Baseline NB	none	.396	.131	.000	.092
M1 Text LogReg	none	.586	.584	.354	.570
M1 Text LogReg	A1	.589	.582	.336	.616
M1 Text LogReg	A2	.586	.593	.383	.567
M1 Text LogReg	A1+A2	.586	.588	.366	.612
M2 Text SVM	none	.591	.612	.405	.621
M2 Text SVM	A1	.587	.604	.399	.623
M2 Text SVM	A2	.590	.614	.417	.614
M2 Text SVM	A1+A2	.587	.609	.417	.620
M3 Text+keyword SVM	none	.934	.924	.845	.496
M3 Text+keyword SVM	A1	.941	.927	.822	.627
M3 Text+keyword SVM	A2	.933	.920	.828	.497
M3 Text+keyword SVM	A1+A2	.943	.930	.832	.620
M4 Hybrid rule+SVM	none	.942	.930	.842	.621
M4 Hybrid rule+SVM	A1	.942	.930	.845	.623
M4 Hybrid rule+SVM	A2	.941	.929	.837	.614
M4 Hybrid rule+SVM	A1+A2	.943	.930	.837	.620

Table 3: All models and augmentation strategies (5-fold grouped cross-validation, mean over folds).

Experiment	Hyperparameter	Values tested	Macro-F1 range	Fold std	Kept
M1 LogReg	C	1, 10, 100	.575–.589	.018–.024	10
M2 SVM	C	0.1, 0.5, 1, 2	.606–.612	.020–.028	0.5
M2 SVM	min. doc. frequency	1, 2, 5	.609–.615	.020–.027	2
M3 SVM+kw (A1)	keyword weight	1, 3, 10	.925–.929	.008–.013	3
M2 + A2	target size	150, 300	.608–.614	.015–.019	150
M2 + A2	deletion probability	0.1, 0.2, 0.4	.606–.614	.014–.020	0.2

Table 4: Hyperparameter grid (5-fold grouped cross-validation). All differences are within one fold standard deviation.

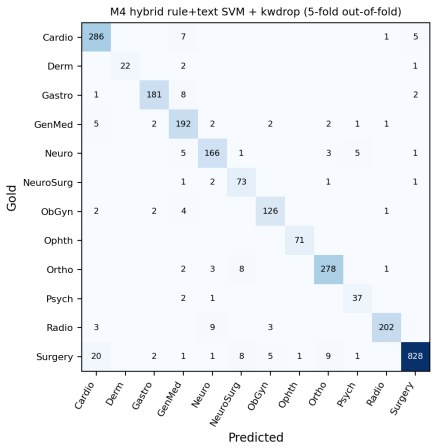


Figure 2: Confusion matrix of the final system (M4 with keyword dropout), out-of-fold predictions.